

Omar Abdul Qadir

Embedded Software Engineer

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Summary:

Determined Embedded Software Engineer with a strong focus on research, problem-solving, and system optimization. Seeking challenging embedded software roles to apply my technical expertise, continue learning, and contribute meaningfully to building reliable and high-quality solutions.

Professional Experience:

Mid-Level Embedded Software Engineer

Pylon

Nov 2023 – Present

- Developed and optimized critical communication stacks, including PLC (PRIME I.3.6), DLMS-HDLC, and TCP/IP over fiber and copper interfaces, ensuring 99% data transmission reliability across smart meter networks.
- Designed and implemented low-level drivers for UART, SPI, and I2C peripherals, supporting rapid data transfer to memory devices (EEPROM, Flash) and communication modules (NFC, RS485).
- Engineered system-level services for memory management, buffering, and storage integrity, successfully preventing data corruption under heavy system loads.
- Architected and transitioned bare-metal super-loop systems to RTOS-based environments, improving system modularity and CPU resource utilization by an estimated 95% Idle.
- Executed hardware bring-up, debugging, and integration of embedded communication interfaces across 30+ custom PCB iterations, accelerating development cycles by 60% while ensuring robust device-level connectivity.
- Architected the core application structure and custom build system for a residential Data Concentrator Unit (DCU), building and deploying full custom Embedded Linux images to seamlessly manage edge data collection for 120+ smart grid nodes.
- Integrated and implemented low-level hardware drivers and system initialization (bring-up) services, engineering critical environment control services including RTC synchronization, and successfully reducing system boot time from 4 minutes (legacy vendor average) to just 45 seconds.

Embedded Systems Part time Instructor

Smart Technology

Jun 2025 – Present

- Authored and structured comprehensive technical curricula and 20+ advanced practical challenges (Mega-Assignments) for the ATmega32 MCU, covering low-level concepts including GPIO, Timers, PWM, and External Interrupts.
- Delivered technical instruction and C/C++ code reviews to nearly 60 students, accelerating their troubleshooting capabilities and bare-metal architecture understanding.
- Designed and evaluated rigorous laboratory projects, including an Input Capture Unit (ICU) task integrating the HC-SR04 ultrasonic sensor, and DC motor control systems emphasizing direction-toggling logic.
- Formulated assessment frameworks and grading criteria to evaluate hardware-software integration projects, significantly enhancing overall engineering readiness.

Courses:

Machine learning for Embedded Systems

Udemy

June 2026

- Deployed end-to-end TinyML applications, focusing on hardware-aware model optimization and neural network inference for the ARM Ethos-U microNPU architecture.
- Integrated machine learning workflows on resource-constrained Edge devices, utilizing Cortex-M systems to achieve optimal performance-per-watt in embedded environments.
- Applied advanced model compression techniques, including weight pruning and clustering, to effectively run memory-bound deep learning models on specialized embedded hardware.

The Complete Prompt Engineering for AI Bootcamp (2026)

Udemy

April 2026

- Mastered advanced prompt design techniques, including zero-shot, few-shot, and chain-of-thought prompting, to optimize Large Language Model (LLM) performance and accuracy.
- Developed strategies for leveraging Generative AI tools in real-world problem-solving, focusing on iterative prompt tuning, role-based prompting, and contextual constraints.
- Acquired expertise in AI ethics and risk mitigation, implementing best practices to reduce model bias, avoid hallucinations, and ensure secure, responsible AI usage.

Embedded Linux

EDGES

September 2024

- Gained hands-on experience with Embedded Linux systems, including YOCTO, Buildroot, and manual build system.
- Developed and deployed Linux applications using C++ with OOP principles, supported by custom build systems (Makefiles and CMake).
- Wrote and optimized Bash scripts for automation, system setup, debugging, and workflow efficiency.
- Built and customized embedded Linux environments, integrating drivers, services, and user-space components while focusing on performance and reliability.

Education:

Bachelor Degree in Systems and Biomedical Engineering

El Shorouk Academy, Cairo, Egypt

2015 – 2020

Core Skills:

Programming Languages: C, C++, Python, Bash.

Communication Protocols: SPI, I2C, UART, CAN, LIN, PLC (Power line communication), Ethernet, Wireless (BLE, WI-FI, Zegbee).

Embedded Systems: RTOS, Embedded Linux, Bare-metal, Device Drivers, Bootloaders, ARM Cortex-M, Firmware Development, Memory Management, Hardware Bring-up.

Tools: Yocto, Buildroot, GDB, JTAG, SWD, Logic Analyzer, Oscilloscope, Git, Makefile, CMake, CI/CD.

Language Skills:

- Arabic: Native speaker.
- English: Advanced level.
- Spanish: Beginner level.
- German: Beginner level.

All references are available upon request.